

Kaiwen Qiu

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EDUCATION

Rutgers University, New Brunswick, NJ	2021–Present
Ph.D. in Econometrics and Quantitative Economics	
Princeton University, Princeton, NJ	2022
Exchange Ph.D. Student	
Georgetown University, Washington, DC	2021
M.S. in Econometrics and Quantitative Economics, Computational Economics Track	
University of Colorado Boulder, Boulder, CO	2019
B.A. in Mathematical Economics	

FIELDS OF INTEREST

Applied Econometrics; Forecasting; Machine Learning; LLM-based Sentiment Analysis; Macroeconometrics; Time Series

RESEARCH

Qiu, K., Kabiri, A., Landon-Lane, J., and Swanson, N. R. (2026). **New Evidence of the Usefulness of Real-Time Sentiment Indices in Econometric and Judgement Based Forecasting Models.** *Job Market Paper; Journal Submitted.*

Abstract: In this paper, we construct a set of new narrative sentiment indices using textual data from a large corpus of U.S. economic and policy documents. The indices are designed for use in real-time forecasting contexts, and are constructed using transformer-based language models and LLM-assisted annotation. They include: a finance-domain transformer; an economics-tone model trained on LLM-generated annotations of Federal Reserve, Bureau of Labor Statistics, and Census text; and a topic-conditioned narrative measure. In order to determine the usefulness of these three indices for forecasting, we carry out a series of experiments in which their marginal predictive content is assessed. In particular, we first assess the predictive content of the indices in econometric models that are used for real-time U.S. macroeconomic and financial forecasting. We then assess whether our sentiment indices contain information that can be utilized to improve the (quasi) judgemental forecasts reported in the United States Survey of Professional Forecasters (SPF). This is done by testing the rationality of SPF forecasts when conditioning on both real-time macroeconomic variables and our sentiment indices. Our prediction experiments indicate that the new sentiment indices offer substantial marginal predictive content for a variety of variables. We also find that information contained in the indices is utilized by participants in the Survey of Professional Forecasters, as evidenced by the paucity of rejections of efficiency in our rationality test experiments, providing evidence that public reporting of the indices is warranted.

Chao, J., Liu, Y., **Qiu, K.**, and Swanson, N. R. (2026). **Consistent Factor Estimation and Forecasting in Factor-Augmented VAR Models.** R&R at *Quantitative Economics*.

Abstract: We introduce a completely consistent method for variable selection with high dimensional datasets. The method is presented in a framework where latent factors are estimated for the purpose of dimension reduction, and is meant to serve as a complement to extant methods. We argue that the method is of particular interest in empirical settings where there may be many irrelevant predictor variables. The reason for this is that situations where there are “too many” irrelevant variables can lead to inconsistent factor estimates. Interestingly, our method yields a consistent estimate of the number of such irrelevant variables, which can aid the applied practitioner in assessing the strength of the underlying factor structure for a particular application. We also show that when factors constructed using our variable selection method are inputted into the forecast equations implied by a factor augmented vector autoregressive (FAVAR) model for the purpose of forecasting, the conditional mean forecast equations can be consistently estimated. Monte Carlo results are presented indicating that the variable selection method performs well in finite samples. The paper also contains two empirical illustrations, where we compare forecasts constructed from factor-augmented forecast equations using our variable selection procedure with two alternative methods for factor construction - the conventional PCA procedure, which does not pre-screen

the variables, and a hard thresholding method that is widely used in the empirical literature. Overall, we find that our method to outperform both of these alternative procedures in a majority of the cases that we study across different target variables, forecast horizons, and data window specifications (i.e., recursive or rolling).

Qiu, K. (2026). Diffusion Indices, Policy Rules, and LLM Narratives: Forecasting the U.S. Yield Curve in Real Time. *Working paper*.

Abstract: Econometricians increasingly rely on machine-learning tools to summarize information in data-rich environments, yet the relative forecasting value of (i) macro-based diffusion indices, (ii) text-based (LLM) sentiment indices, and (iii) monetary-policy-rule information remains unclear within a unified term-structure framework. This paper conducts a horse race across these information sets by embedding them into a dynamic Nelson–Siegel class model, including its rotated short-rate-based representation (RDNS). We compare three families of real-time diffusion indices constructed from large macro panels: principal components (PCA), targeted shrinkage via LASSO/Elastic Net (EN), and the Chao–Swanson (CS) variable-selection approach. We further introduce three LLM-based text indices designed to capture policy-relevant narratives: an off-the-shelf FinBERT sentiment measure, an economics-domain trained BERT index, and a topic-fusion fine-tuned index that integrates topic structure with supervised sentiment signals. Finally, we incorporate monetary policy information through Taylor-rule components and Taylor-rule residuals (interpreted as policy shocks), motivated by the forward-looking policy-rule literature. Using a real-time rolling-window design, we forecast yields across maturities and horizons. Our results show that (i) CS-based diffusion indices extract more forecasting-relevant macro information than conventional LASSO/EN targeted-PCA constructions, (ii) LLM-based indices contain incremental predictive content beyond macro diffusion indices, particularly around policy-relevant episodes, and (iii) adding Taylor-rule components and/or policy shocks improves short- and intermediate-horizon yield forecasts, with the largest gains concentrated at the short end of the curve. Overall, combining policy-rule information with data-rich indices delivers the most reliable improvements in predictive accuracy for the U.S. Treasury yield curve.

Long, L., Zhang, C., Ma, C., Wang, H., Ji, L., Gao, F., Zhang, J., and **Qiu, K. (2026). Algorithmic Trading by Reinforcement Learning in a Collaborative Manner. *Applied Soft Computing*, 197, 115168.**

Abstract: The application of reinforcement learning approaches in algorithmic trading has been studied widely in recent years. The majority of research concentrates on processing time series and refining neural network architectures, particularly within the framework of single-agent approaches. Some studies have demonstrated that employing multi-agent or ensemble learning can enhance performance. However, the prevalent practice of utilizing ensemble neural networks or various reinforcement learning methods to train simultaneously on the entire time series results in increased computational costs and inefficient recognition of market fluctuations. In this paper, we address these challenges by dividing the lengthy time series data into shorter segments, and introducing a collaborative training framework akin to distributed learning to train multiple clients on these segments. A self-imitation method is integrated to improve collaboration among these clients. Additionally, we propose a multi-loss function that combines the strengths of reinforcement learning with those of imitation learning. Building on this foundation, a Collaborative Self-Imitation Double Deep Q Network (CSI-DDQN) is proposed to train these clients effectively. Our experimental results, conducted using a multi-layer perceptron (MLP) architecture on U.S. stock data, demonstrate that our proposed method significantly outperforms traditional baseline models, particularly excelling in the performance metric of Sharpe ratio (SR). This highlights its effectiveness in maximizing returns and mitigating risks. Furthermore, to demonstrate the flexibility and adaptability of the newly proposed framework to different market conditions, more experiments were conducted by replacing the corresponding parts in the current framework with other compute plans, network

WORK IN PROGRESS

Hard and Soft Information in Volatility Forecasting: Latent Factors and LLM Sentiment. With Yang Liu

This paper combines latent volatility factors from high-frequency stock data with language model based news sentiment to forecast daily equity market volatility. It examines whether hard price-based and soft narrative-based information offer complementary predictive content both jointly and individually.

Integrating Real-Time Survey Information into NLP and LLM Sentiment Indices.

Economic narratives and sentiment surveys offer distinct perspectives on macroeconomic conditions. This project explores whether integrating real-time survey information into language model based sentiment measures improves macroeconomic forecasting and helps identify information not fully reflected in professional forecasts.

WORK EXPERIENCE

Quantitative Researcher Intern, PGIM, Newark, NJ June–Aug. 2025

- Developed FX timing signals using nonlinear variable selection and random forests; backtested long–short strategies.
- Used SHAP values for feature attribution to support interpretable risk budgeting.
- Implemented transfer-entropy signals linking daily Economic Surprise Index shocks to intraday FX movements.

Research Assistant, Rutgers University, New Brunswick, NJ May 2024–May 2025

- Worked on the `DrddidGmm` R package, implementing the doubly robust GMM estimator in Xu (2024) for average treatment effects on the treated and spillover effects in difference-in-differences designs, with heteroskedasticity-, cluster-, and spatial-robust standard errors.

Research Assistant, Rutgers University, New Brunswick, NJ March–May 2024

- Collected TAQ intraday data for 380 S&P 500 constituents and the SPDR S&P 500 ETF (SPY).
- Applied Chao–Swanson variable selection and shrinkage methods to select stocks and features, including estimated factors, with forecast-based cross-validation for tuning.

Research Assistant, Rutgers University, New Brunswick, NJ March–Dec. 2023

- Constructed first-release real-time datasets for more than 130 macroeconomic and financial variables from FRED.
- Built an emotion measure from Reuters Historical News Archive text, compared it with the relative shift sentiment (RSS) lexicon, and evaluated its incremental predictive content in judgmental and econometric forecasts.

IT Summer Associate, Thermo Fisher Scientific, Shanghai, China June–Aug. 2019

- Contributed to an order-automation project using Automagica, an open-source Python platform for robotic process automation.

Summer Risk Analyst, China Pacific Insurance, Shanghai, China June–Aug. 2018

- Organized critical illness insurance policy data and supported analysis of urban residents’ insurance coverage and the construction and validation of risk models.

TEACHING EXPERIENCE

Instructor, Rutgers University 2024–2026

- Econometrics (ECON 322) Summer 2024; course evaluation: 4.71
- R Workshop Summers 2024, 2025, 2026

Teaching Assistant, Rutgers University 2022–2025

- Economics for Big Data (master’s level) Spring 2025
- Advanced Time Series Spring 2025
- Econometrics II (Ph.D. level) Fall 2024
- Econometrics Fall 2023, Spring 2024, Fall 2024
- Econometrics I (Ph.D. level) Spring 2024
- Introduction to Macroeconomics Spring 2023
- Money, Banking and the Financial System Spring 2023
- Intermediate Macroeconomic Analysis Fall 2022

Undergraduate Economics Tutor, University of Colorado Boulder Spring 2018

REFeree SERVICE

Journal of Time Series Analysis (2); *International Journal of Forecasting* (3); *Journal of Financial Econometrics* (1)

CONFERENCE PRESENTATIONS

Midwest Economics Association (MEA) April 2026

“Diffusion Indices, Policy Rules, and LLM Narratives: Forecasting the U.S. Yield Curve in Real Time”

“Consistent Factor Estimation and Forecasting in Factor-Augmented VAR Models, with an Application to Forecasting Stock Returns”

HONORS AND AWARDS

• Graduate Assistantship, Rutgers University	2025–2026
• Prof. Hiroki Tsurumi and Peter J. Zangari Graduate Dissertation Award, Rutgers University	Spring 2025
• Peter Asch Memorial Scholarship, Rutgers University	Spring 2025
• Sidney I. Simon Teaching Award, Rutgers University	Spring 2025
• Honorable Mention, Sidney I. Simon Prize, Rutgers University	Spring 2024
• Sidney Grant, Rutgers University	Spring 2024
• Sidney Brown Prize in Economics, Rutgers University	Spring 2023
• Teaching Assistantship, Rutgers University	2022–2025
• Dean’s List, University of Colorado Boulder	Spring 2018

SKILLS

Programming and statistical software: Python, R, MATLAB, Stata, GAUSS

Languages: English (fluent), Mandarin (native)

REFERENCES

Norman R. Swanson (Chair)

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Rutgers University
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